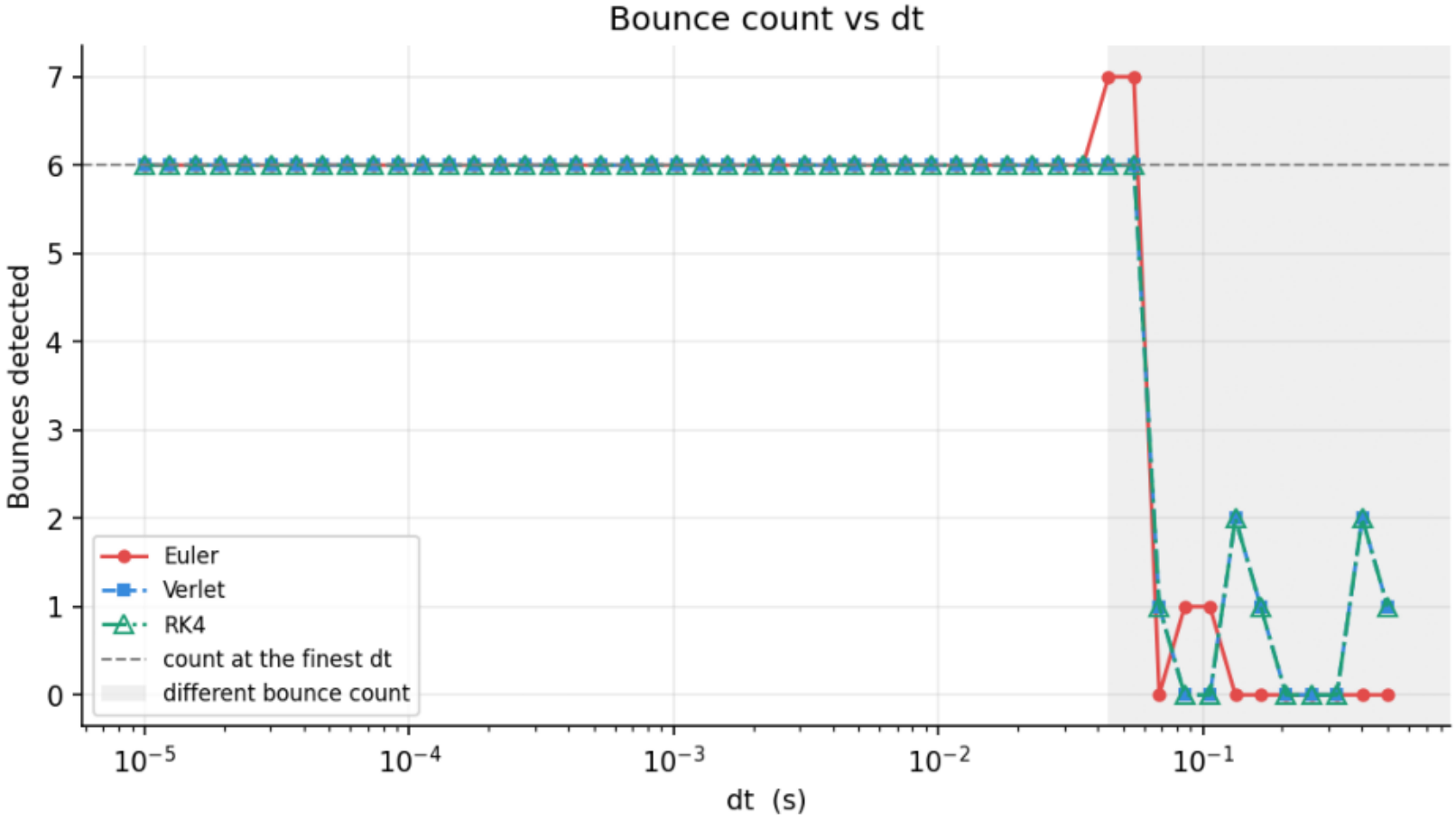
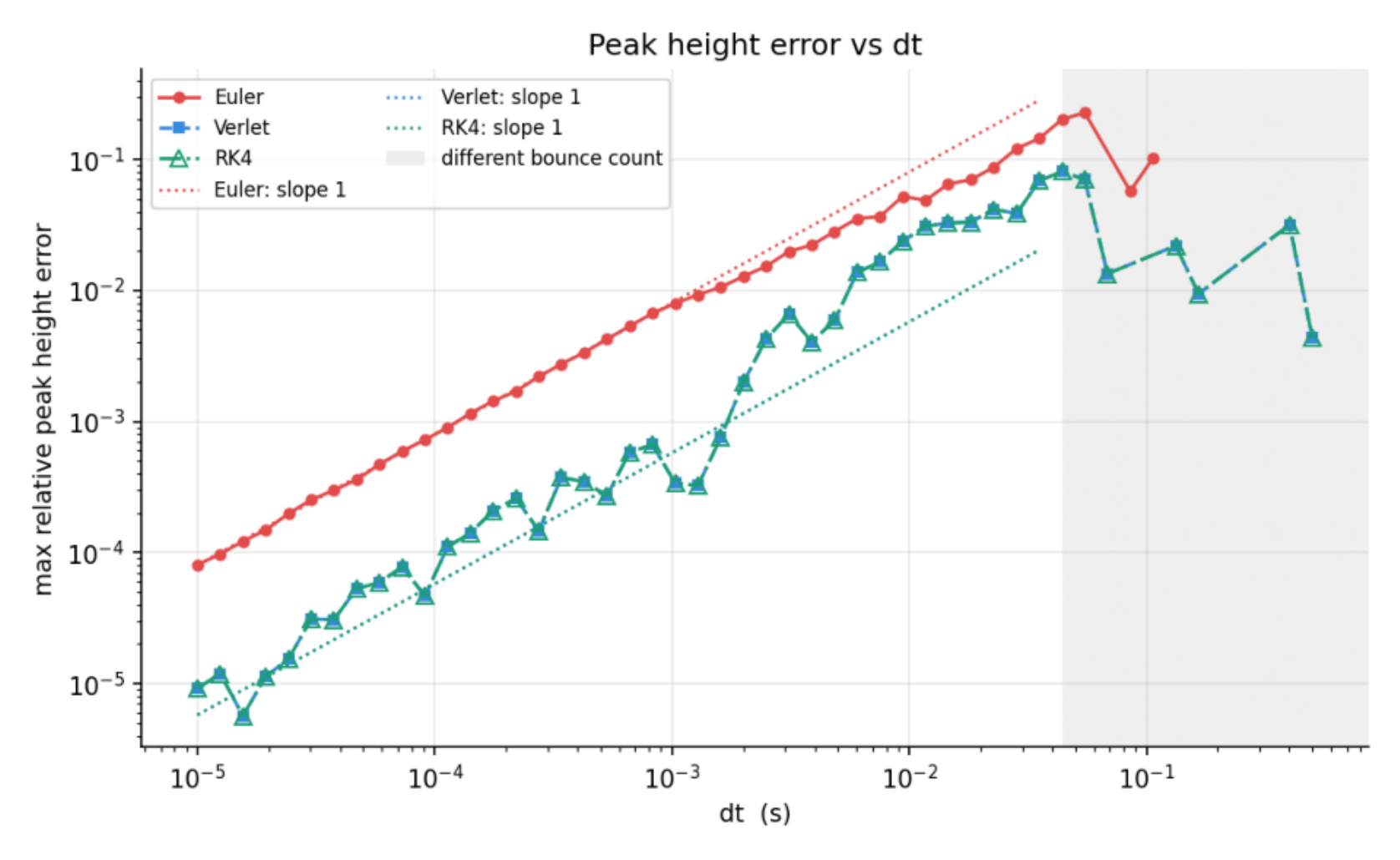
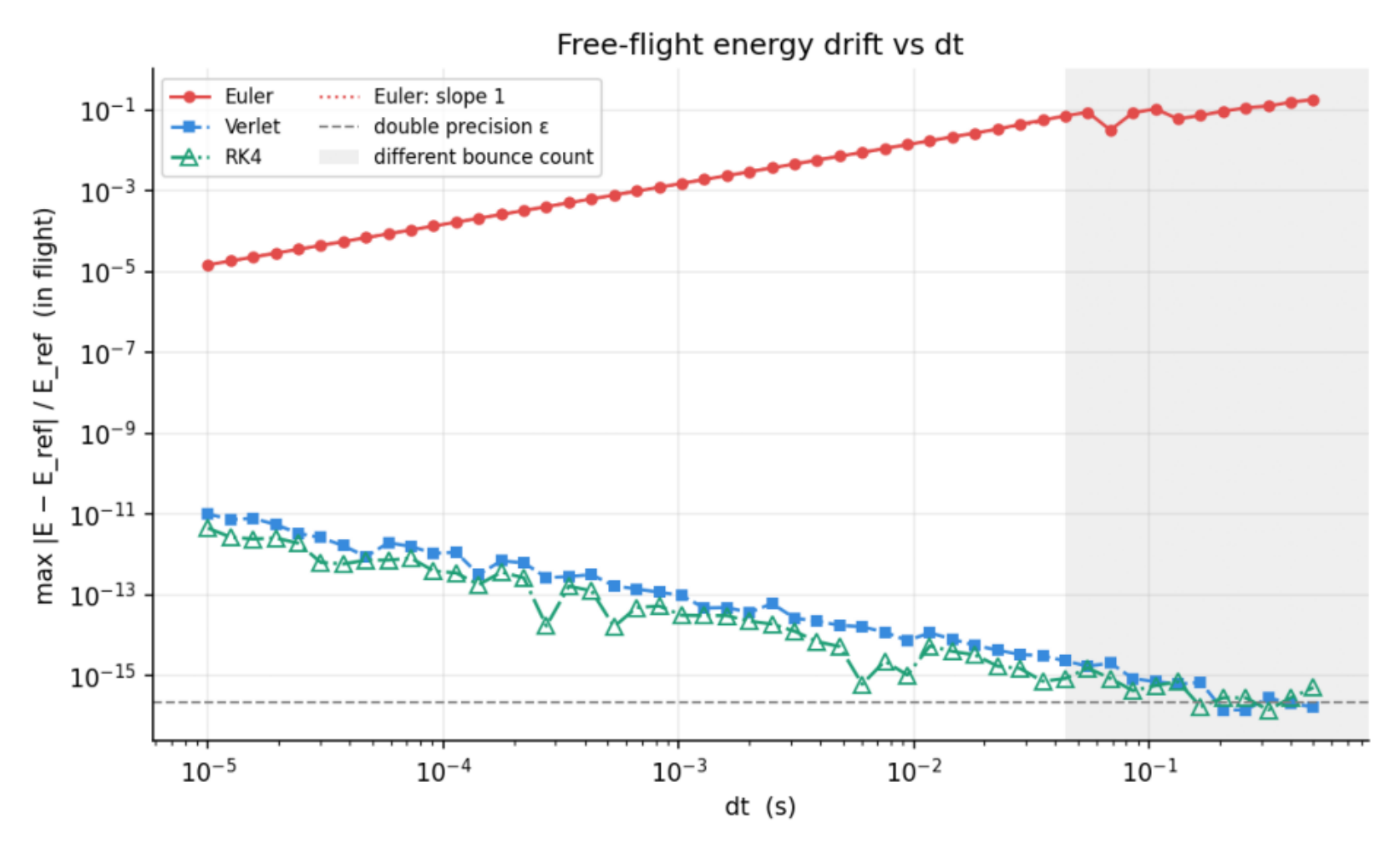
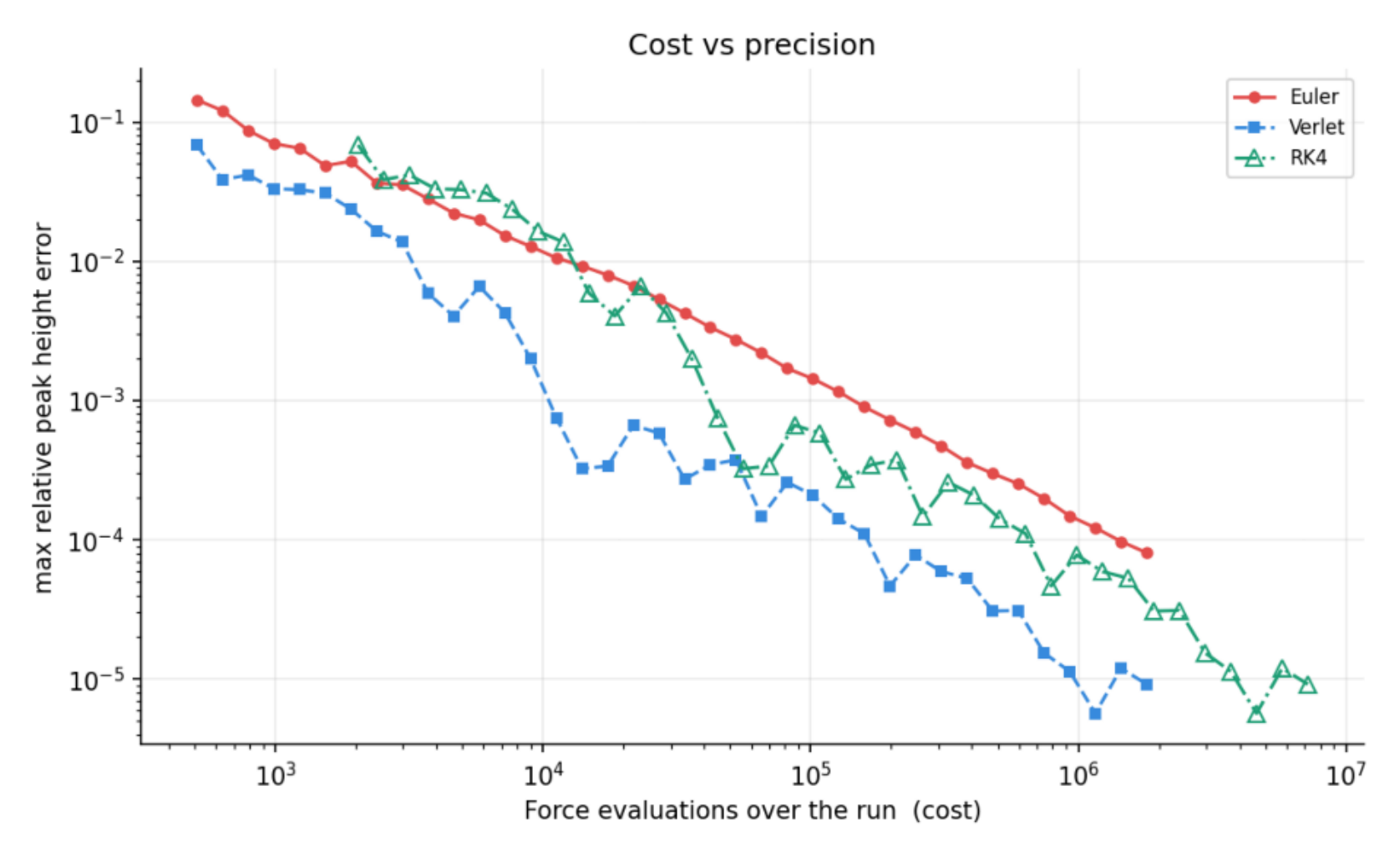
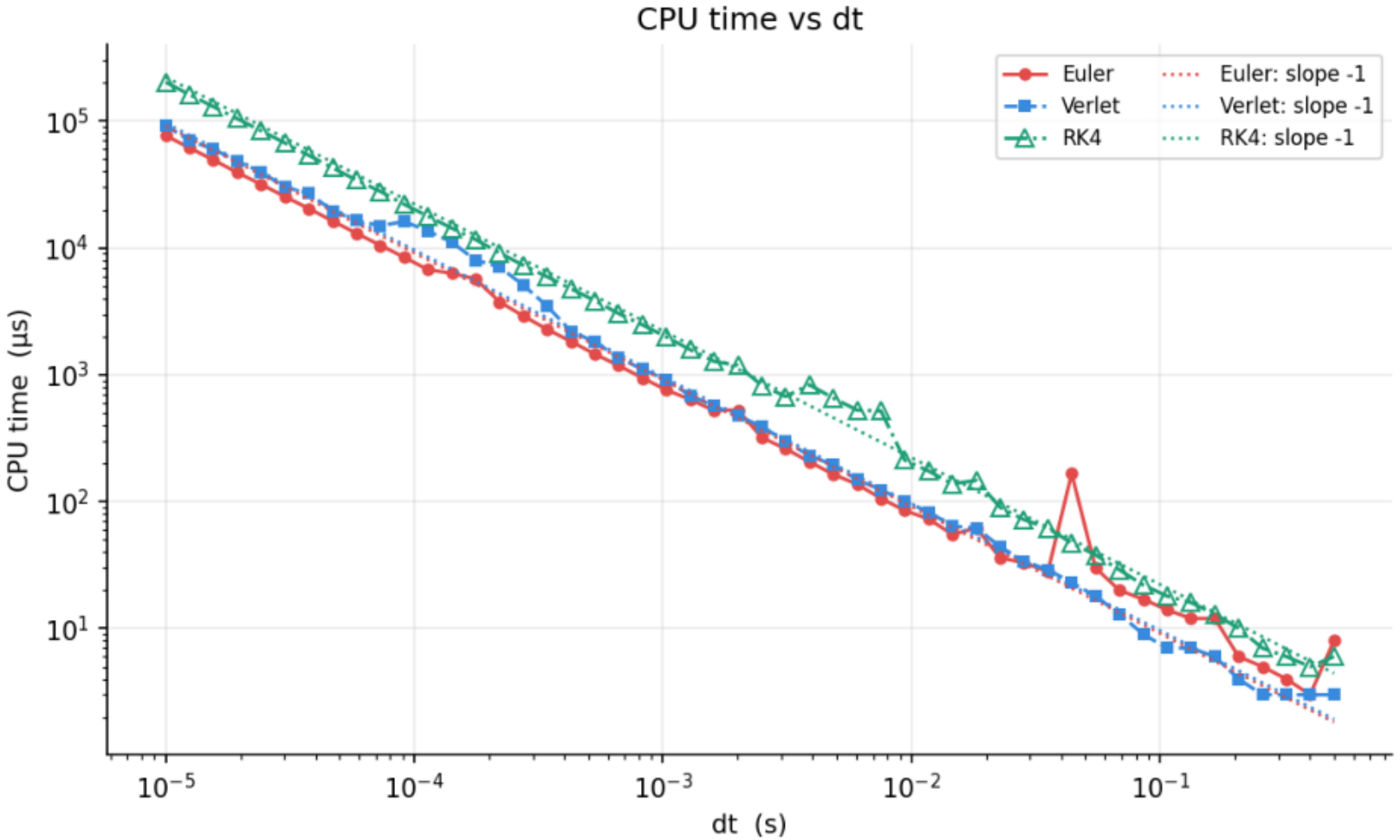
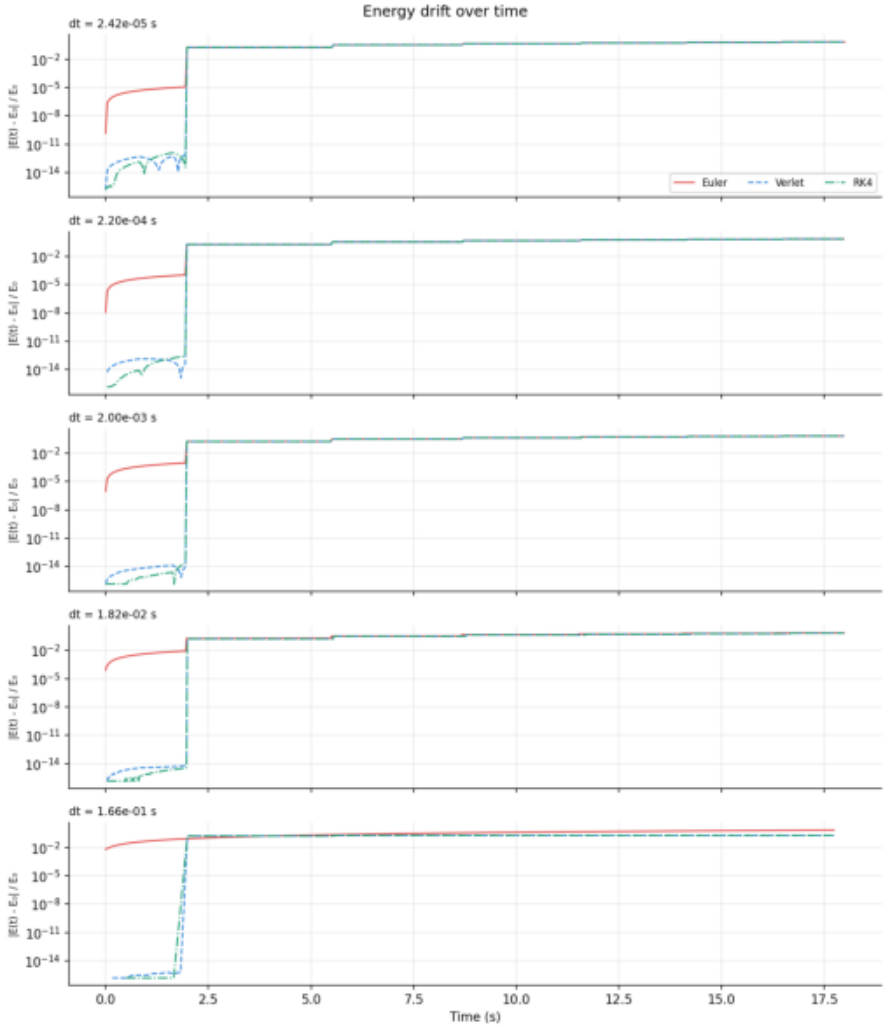

Bounce count vs dt
Bounces detected
dt (s)
Euler
Verlet
RK4
count at the finest dt
different bounce count


Peak height error vs dt
Euler
Verlet
RK4
Euler: slope 1
Verlet: slope 1
RK4: slope 1
different bounce count
max relative peak height error
dt (s)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}


Free-flight energy drift vs dt
Euler
Verlet
RK4
Euler: slope 1
double precision ε
different bounce count
max |E − E_ref| / E_ref (in flight)
dt (s)
10^{-1}
10^{-3}
10^{-5}
10^{-7}
10^{-9}
10^{-11}
10^{-13}
10^{-15}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}


Cost vs precision
Euler
Verlet
RK4
max relative peak height error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
Force evaluations over the run (cost)


CPU time vs dt
Euler
Verlet
RK4
Euler: slope -1
Verlet: slope -1
RK4: slope -1
CPU time (µs)
dt (s)
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}


Energy drift over time
dt = 2.42e-05 s
dt = 2.20e-04 s
dt = 2.00e-03 s
dt = 1.82e-02 s
dt = 1.66e-01 s
|E(t) - E₀| / E₀
Euler
Verlet
RK4
Time (s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5